## SECTION-I

### Single Correct Answer Type 3 Q. [3 M (–1)]

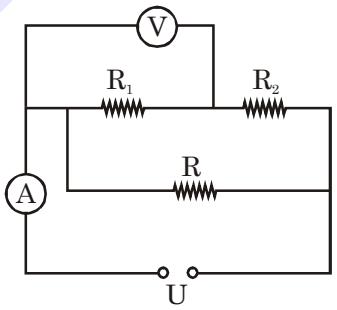
**1.** A micrometer has a resistance of 10 Ω and a full scale range of 50 μA. It can be used as a voltmeter or as a higher range ammeter provided a resistance is added to it. The correct range and resistances combination (approximately) :-
(A) 50 V range with 10 kΩ resistance in series (B) 10 V range with 200 kΩ resistance in series
(C) 5 mA range with 1Ω resistance in parallel (D) 10 mA range with 1 Ω resistance in parallel

**2.** The resistance of a galvanometer is normally measured using :
(A) Post office box (B) Potentiometer
(C) Half deflection method (D) Meter bridge

**3.** In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when:
[1] battery emf is increased [2] all resistances are increased by 10 ohms
[3] all resistances are made five times [4] the battery and the galvanometer are interchanged
(A) only [1] is correct (B) [1], [2] and [3] are correct
(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

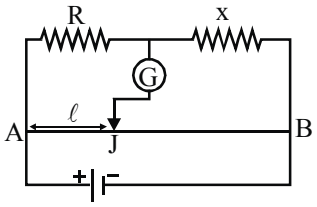
### Multiple Correct Answer Type 7 Q. [4 M (–1)]

**4.** What are the resistances of the resistors $R_1$ and $R_2$, shown in the figure, if the voltage supply is 40 V, and the power dissipated in resistor R is 80 W, the readings on the ammeter and on the voltmeter are 3 A and 30 V, respectively. (All devices are ideal)




(A) $R_1 = 30\ \Omega$
(B) $R_2 = 30\ \Omega$
(C) $R_1 = 10\Omega$
(D) $R_2 = 10\Omega$

**5.** In the meter bridge experiments initially balance point is J, where AJ=$\ell$. One wants to make some changes in the shown setup. After any change balance length being AJ' = $\ell$', then choose the correct statement(s). [Here R ≠x]



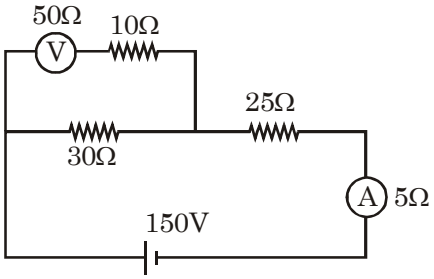

(A) If in shown set-up the galvanometer and battery are inter changed current through galvanometer is zero.
(B) radius of wire AB is double then $\ell' = 2\ell$
(C) if length of wire AB is doubled then $\ell'=2\ell$
(D) If the values of R and x both are double and then interchanged then $\ell'=\ell$

**6.** In the standard post office box setup
(A) the null point reading implies the position of jockey for which reading of galvanometer is zero.
(B) the null point reading means consecutive readings at which the galvanometer deflects in the opposite direction.
(C) the key connected to Galvanometer is switched on after switching on the key connected to battery.
(D) the key connected to Galvanometer is switched off after switching off the key connected to battery.

**7.** A 20 m long potentiometer wire has a resistance of 20 ohm. It is connected in series with a battery of emf 3V and a resistance of 10Ω. The internal resistance of cell is negligible. If the length can be read accurately up to 1 mm, the potentiometer can read voltage
(A) up to minimum of 0.2 mV (B) with an accuracy of 0.2 mV
(C) with an accuracy of 0.1 mV (D) upto maximum of 2V


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**8.** For given circuit, choose the correct option(s) :



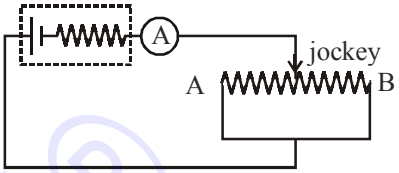

(A) Reading of voltmeter is 50 V

(B) Reading of voltmeter is 60 V

(C) Reading of ammeter is 3A

(D) Reading of ammeter is 4A

**9.** The circuit diagram shows a source of emf 10 V having an internal resistance of 2.5Ω connected to an ideal ammeter. The jockey can be slided to different points on the Rheostat whose total resistance is 10Ω. What can be the reading of the ammeter.




(A) 1A (B) 2A

(C) 3A (D) 4A

**10.** Two identical moving coil galvanometer have 10 Ω resistance and full scale deflection at 2 μA current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with R = 1000 Ω resistor by using an ideal cell. Which of the following statement(s) is/are correct ?

(A) The measured value of R will be 978 Ω < R < 982 Ω.

(B) The resistance of the Voltmeter will be 100 kΩ.

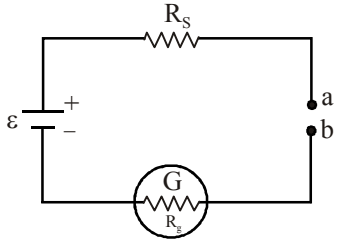
(C) The resistance of the Ammeter will be 0.02 Ω ( round off to 2$^{nd}$ decimal place)

(D) If the ideal cell is replaced by a cell having internal resistance of 5 Ω then the measured value of R will be more than 1000 Ω.

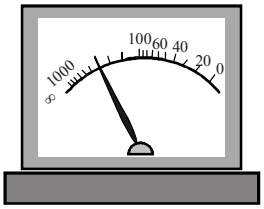
## Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
## (Single Correct Answer Type)

**Paragraph for question nos. 11 to 13**

A simple ohmmeter consists of battery connected in series with a galvanometer and a resistor, as shown in figure. The resistance $R_s$ is chosen such that when the terminals a and b are shorted (put in electrical contact, with negligible resistance between them), the current through the galvanometer gives a full-scale deflection. Thus, a full scale deflection indicates no resistance between terminals a and b. A zero deflection indicates an infinite resistance between the terminals. When the terminals are connected across an unknown resistance R, the current through the galvanometer depends on R, so the scale can be calibrated to give a direct reading of R, as shown in figure. Because an ohmmeter sends a current through the resistance to be measured, some caution must be exercised when using this instrument. For example, you would not want to try to measure the resistance of sensitive ammeter with an ohmmeter, because the current provided by the battery in the ohmmeter would probably damage the ammeter. Let us use a galvanometer with resistance of 20 Ω and maximum current of 10 mA, ε = 1 V. For a current more that 10 mA, galvanometer would be damaged.



(a)



(b)


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**11.** The scale is non linear because
(A) the resistance changes with change in temperature.
(B) of internal resistance of the battery.
(C) the current in circuit decreases with increase in resistance.
(D) of internal resistance of galvanometer.

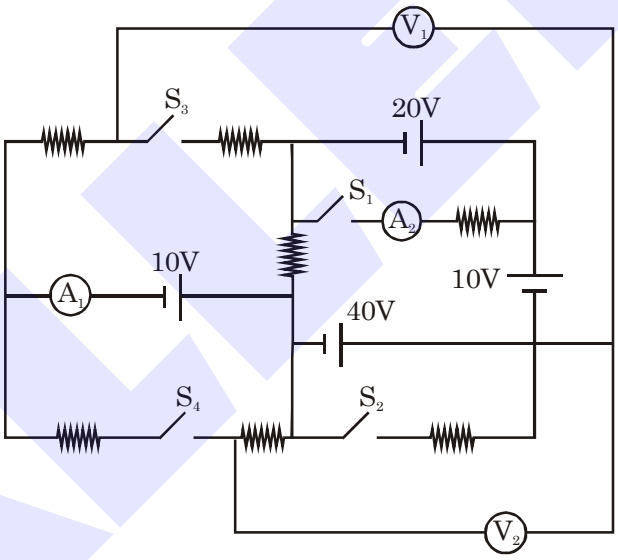
**12.** The minimum resistance required for rest of the circuit (other than galvanometer) is
(A) 20Ω (B) 30Ω (C) 80Ω (D) 100Ω

**13.** The reading in galvanometer when resistance connected between a and b is 100Ω is
(A) 5 mA (B) 3.3 mA (C) 2mA (D) 8mA

**Matching list based comprehension Type (4 × 4 × 4) 1 Table × 3 Q. [3(–1)]**
**Single option correct (Three Columns and Four Rows)**

**Answer Q.14, Q.15 and Q.16 by appropriately matching the information given in the three columns of the following table.**

In given circuit, all batteries, voltmeters and ammeters are ideal. Voltmeters and ammeters have only +ive scale for reading. Resistance of each resistor is 5Ω. Reading of voltmeters are $V_1$ and $V_2$ and of ammeters are $A_1$ and $A_2$. Initially all switches are closed.




| | Coloumn–I $\lvert A_1 - A_2 \rvert$ | | Column-II $V_1$ | | Column-III $V_2$ |
|---|---|---|---|---|---|
| (I) | 3 A | (i) | 20 V | (P) | 45 V |
| (II) | 0 A | (ii) | 30 V | (Q) | 35 V |
| (III) | 5 A | (iii) | 50 V | (R) | 40 V |
| (IV) | 1 A | (iv) | 40 V | (S) | 25 V |

**14.** Only switch $S_2$ is open
(A) (I) (iii) (S) (B) (IV) (ii) (P) (C) (I) (iii) (P) (D) (IV) (ii) (S)

**15.** Only switch $S_3$ is open :
(A) (III) (iii) (R) (B) (IV) (ii) (S) (C) (I) (iii) (P) (D) (II) (iv) (Q)

**16.** Only switch $S_1$ is open :
(A) (III) (iii) (P) (B) (I) (iii) (R) (C) (IV) (ii) (P) (D) (III) (ii) (P)

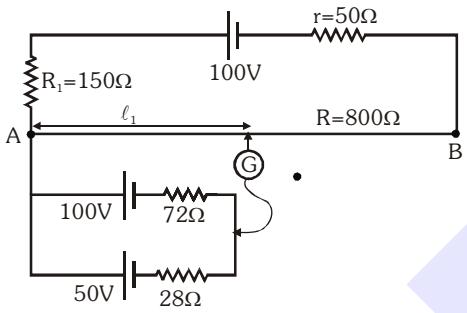

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## SECTION-II

**Numerical Answer Type Question** **4Q.[3(0)]**

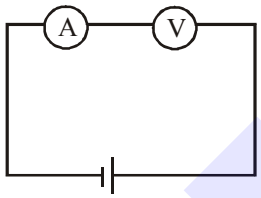
**(upto second decimal place)**

**17.** In the circuit shown length of A is 100 cm. Then find the value of $\ell_1$ (in cm). Where $\ell_1$ corresponds to null point.

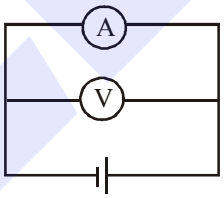



**18.** The deflection of a moving coil galvanometer falls from 60 divisions to 12 divisions for the same value of current in the circuit, when a shunt of 12Ω is connected. What is the resistance (in ohm) of the galvanometer ?

**19.** In circuit (a) the instruments read $U_1 = 100$ V, $I_1 = 5\mu$A. In circuit (b) we have $U_2 = 25$ mV, $I_2 = 2.5$ A. What is the internal resistance of the cell (in Ohm). Round off to nearest integer. Consider both instruments to be regular laboratory instruments.

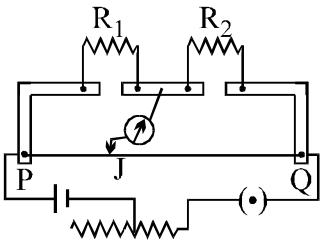


(a)



(b)

**20.** The circuit diagram given in the figure shows the experimental setup for the measurement of unknown resistance by using a meter bridge. The wire connected between the points P & Q has non-uniform resistance such that resistance per unit length varies directly as the distance from the point P. Null point is obtained with the jockey J with $R_1$ and $R_2$ in the given position. On interchanging the positions $R_1$ and $R_2$ in the gaps the jockey has to be displaced through a distance $\Delta$ from the previous position along the wire to establish the null point. If the ratio of $\frac{R_1}{R_2} = 3$, find the value of $\Delta$ (in cm). Ignore any end corrections. [Take $\sqrt{3} = 1.7$]



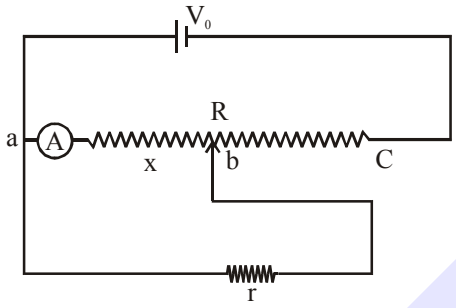


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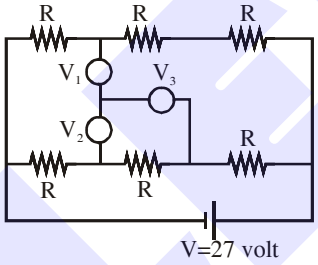
## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]**

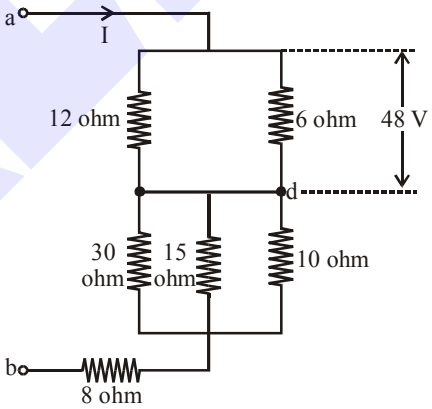
**21.** A constant voltage $V_0$ (= 12V) is applied to a potential divider of resistance R (= 4Ω), connected to an ideal ammeter. A constant resistor r (= 2 Ω) is connected to the sliding contact of the potential divider (as shown). Find the minimum current (in A) measured by ammeter.




**22.** In the circuit shown below, all the voltmeter identical and have very high resistance. Each resistor has the same resistance. The voltage of the ideal battery shown is 27 V. Find the reading of voltmeter $V_3$ (in volts).



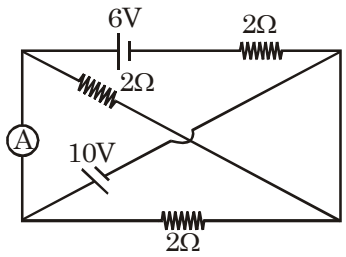

**23.** The potential difference across the 6 ohm resistance in the given circuit is 48 V. The potential difference across a to b is 51 × α Volt. Then find the value of α.




**24.** In a meter bridge experiment, the unknown resistance was connected in left branch and a resistance box in right branch. The resistance in resistance box is 50Ω (known acurately). There is no end correction. The least count of meter bridge is 1mm. The null point was obtained at 40.0cm from left end. Now the radius of wire was measured by a screw guage with pitch of 1mm and no. of divisions on circular scale as 100. The screw guage read 1 main scale division and 60 circular scale divisions. The length of resistor was measured using a vernier callipers with 1mm as 1 main scale division and 9 main scale division = 10 vernier scale divisions. The length is 2.2 cm as main scale reading and $5^{th}$ vernier scale division coinciding. The percentage error in measurement of resistivity is approximately.


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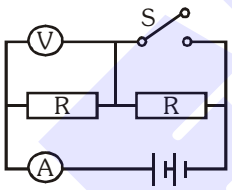
**25.** An ideal ammeter is connected in a circuit as shown in circuit diagram. What will be the reading of ammeter (in S.I. units).




## SECTION-IV

### Matrix Match Type (4 × 5) 1 Q. [8 M (for each entry +2(0))]

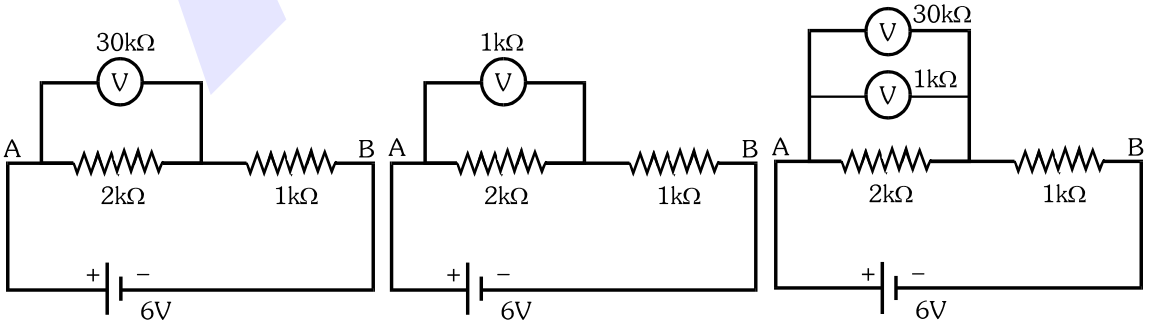
**26.** The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed ?




| Column I | Column II |
|---|---|
| (A) Ammeter reading | (P) Increases |
| (B) Voltmeter reading | (Q) Decreases |
| (C) Equivalent resistance of circuit | (R) Does not change |
| (D) Power dissipated across R in right branch | (S) Becomes zero |
| | (T) Cannot be determined |

### Subjective Type 4 Q. [4 M (0)]

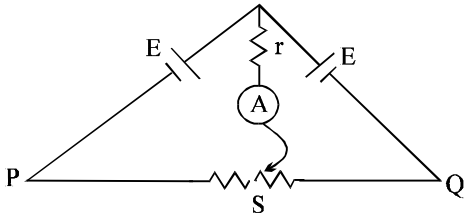
**27.** A series combination of a 2kΩ resistor and a 1kΩ resistor, is connected across a battery of emf 6V and negligible internal resistance. The potential drop, across the 2kΩ resistor, is measured by (a) a 30kΩ voltmeter (b) a 1kΩ voltmeter and (c) both these voltmeters connected across it. If the voltmeter readings in the three cases are $V_1$, $V_2$ and $V_3$ respectively, arrange these readings in descending other.



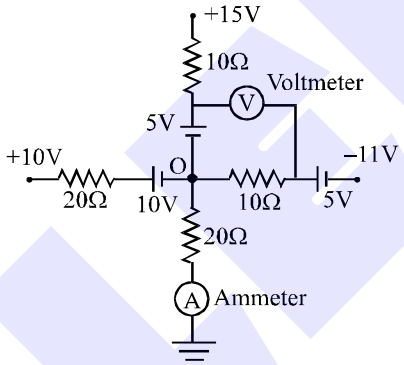

How will the three readings compare with one another if the potential drop were measured across the series combination of the 2kΩ and the 1kΩ resistor i.e., across the points A and B ?


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**28.** In the figure. PQ is a wire of uniform cross-section and of resistance $R_0$. A is an ideal ammeter and the cells are of negligible resistance. The jockey J can freely slide over the wire PQ making contact on it at S. If the length of the wire PS is $f = 1/n^{th}$ of PQ, find the reading on the ammeter. Find the value of 'f' for maximum and minimum reading on the ammeter.




**29.** The potential of certain points in the circuit are maintained at the values indicated. The Voltmeter and Ammeter are ideal. Find the potential of the cross junction point in the circuit (at center O) and the readings of Voltmeter and Ammeter. All cells are ideal.




**30.** A galvanometer (coil resistance 99Ω) is converted into a ammeter using a shunt of 1 Ω and connected as shown in the figure (i). The ammeter reads 3A. The same galvanometer is converted into a voltmeter by connected a resistance of 101Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be 4/5 of the full scale reading. Find

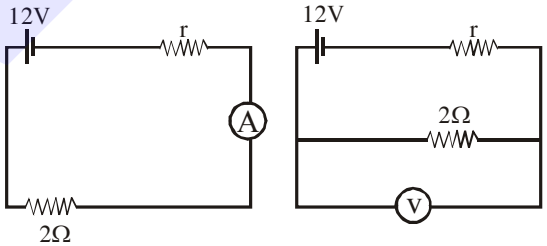



Fig. (i) Fig. (ii)

(i) internal resistance r of the cell
(ii) range of the ammeter and voltmeter
(iii) full scale deflection current of the galvanometer


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**GR # ELECTRICAL INSTRUMENTS** **ANSWER KEY**

## SECTION-I

**Single Correct Answer Type** **3 Q. [3 M (–1)]**

**1. Ans. (B)** **2. Ans. (C)** **3. Ans. (C)**

**Multiple Correct Answer Type** **7 Q. [4 M (–1)]**

**4. Ans. (A,D)** **5. Ans. (AC)** **6. Ans. (B,C)** **7. Ans. (C, D)** **8. Ans. (A,C)**

**9. Ans. (B,C,D)** **10. Ans. (A,C)**

**Linked Comprehension Type (1 Para × 3Q.)** **[3 M (-1)]**

**(Single Correct Answer Type)**

**11. Ans. (C)** **12. Ans. (C)** **13. Ans. (A)**

**Matching list based comprehension Type (4 × 4 × 4)** **1 Table × 3 Q. [3(–1)]**

**Single option correct** **(Three Columns and Four Rows)**

**14. Ans. (B)** **15. Ans. (C)** **16. Ans. (D)**

## SECTION-II

**Numerical Answer Type Question** **4Q.[3M (0)]**

**(upto second decimal place)**

**17. Ans. 80.00** **18. Ans. 48.00** **19. Ans. 40.00** **20. Ans. 35.00**

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **5 Q. [4 M (0)]**

**21. Ans. 2** **22. Ans. 6** **23. Ans. 4** **24. Ans. 2** **25. Ans. 7**

## SECTION-IV

**Matrix Match Type (4 × 5)** **1 Q. [8 M (for each entry +2(0)]**

**26. Ans.** (A) → (P) ; (B) → (P) ; (C) → (Q) ; (D) → (Q,S)

**Subjective Type** **4 Q. [4 M (0)]**

**27. Ans.** First case $V_1 > V_2 > V_3$; Second case $V_1 = V_2 = V_3$

**28. Ans.** $\frac{\varepsilon}{r + R_0(f - f^2)}$ ; for $I_{max}$, f = 0, 1; $I_{min}$, f = 1/2

**29. Ans.** $x = \frac{4}{3}V, 12\frac{1}{3}V, \frac{1}{15}A$

**30. Ans.** (i) 1.01Ω (ii) 0-5 A, 0-10V, (ii) 0.05 A



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